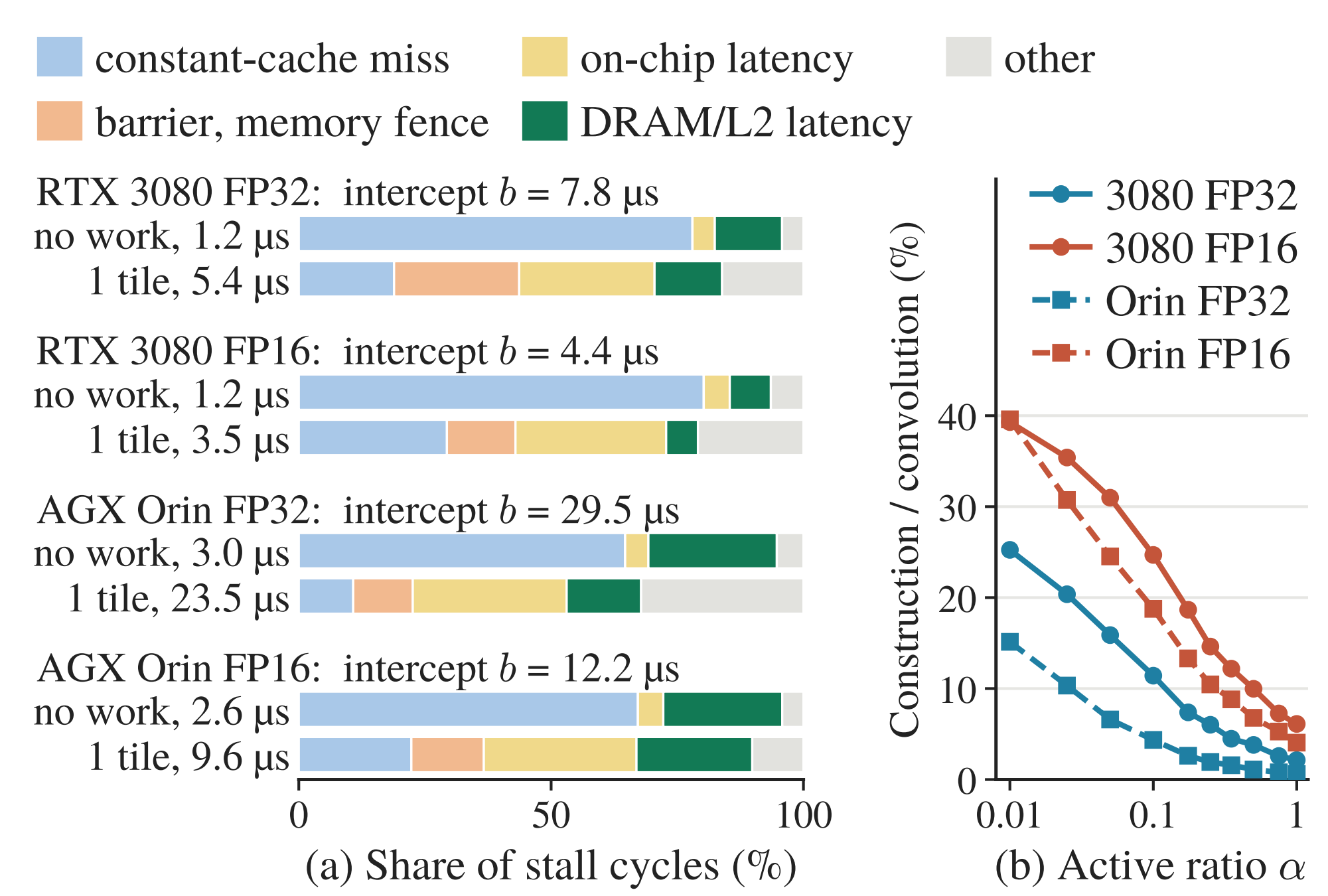
constant-cache miss
on-chip latency
other
barrier, memory fence
DRAM/L2 latency
RTX 3080 FP32: intercept b = 7.8 µs
no work, 1.2 µs
1 tile, 5.4 µs
RTX 3080 FP16: intercept b = 4.4 µs
no work, 1.2 µs
1 tile, 3.5 µs
AGX Orin FP32: intercept b = 29.5 µs
no work, 3.0 µs
1 tile, 23.5 µs
AGX Orin FP16: intercept b = 12.2 µs
no work, 2.6 µs
1 tile, 9.6 µs
0
50
100
(a) Share of stall cycles (%)
3080 FP32
3080 FP16
Orin FP32
Orin FP16
Construction / convolution (%)
40
30
20
10
0
0.01
0.1
1
(b) Active ratio α